

Olympiad Test Paper — Science (Class 7) — Paper 4

Chapter 3: Fibre to Fabric

Paper 4 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	3 — Fibre to Fabric
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic traps in this paper: confusing what each processing step CAUSES versus merely CORRELATES with, reasoning backward from a fibre's burn/wet/shrink behaviour to its protein-vs-cellulose chemistry, mis-locating the single silk-spinning event in the life cycle, conflating sericin with fibroin or guard hairs with undercoat, and attaching a true-sounding but wrong reason to a correct fact. Do not write on this paper — mark answers on your answer sheet.

1. Four unlabelled threads give these flame results: P chars and smells of burnt hair; Q burns fast with a papery smell and grey ash; R chars with a burnt-hair smell; S melts into a hard bead and gives no hair or paper smell. A student concludes 'exactly two of these are animal fibres.' Is the conclusion safe?

- (A) No — three are animal fibres, because Q's grey ash also indicates a protein fibre
- (B) No — only one is an animal fibre, because two threads cannot share the same burnt-hair smell
- (C) Yes — P and R both smell of burnt hair, the signature of protein (animal) fibres, while Q is cellulose (plant) and S behaves like a melting synthetic, so exactly two are animal fibres
- (D) No — none can be called animal fibres from burning alone, because both wool and cotton smell of burnt hair

2. A fabric weakens noticeably when soaked in water, burns with a papery smell, and does not felt or shrink in hot water. A second fabric keeps its strength when wet, burns with a burnt-hair smell, and mats in hot water. A label calls the FIRST one 'pure silk.' The honest verdict is:

- (A) The label is wrong — weakening when wet plus a papery burn smell are cotton (plant) traits, so the first fabric is cotton, not silk
- (B) The label is right — silk weakens when wet and burns with a papery smell, so the first fabric is genuine silk
- (C) Both fabrics are silk, because both are natural fibres that burn
- (D) The label is right and the second fabric must be a synthetic, because it does not weaken when wet

3. Wool, silk and cotton are each set alight. Wool and silk both smell of burnt hair, yet they are NOT the same substance. The single statement that captures BOTH the similarity and the difference is:

- (A) Wool and silk smell alike because they are the identical protein, and cotton differs only because it is dyed
- (B) Wool is keratin and cotton is fibroin, while silk is the cellulose one that smells of paper
- (C) All three are proteins, so all three should smell of burnt hair, and cotton's papery smell is an error of the test
- (D) All three differ in detail, but wool and silk are both proteins (keratin and fibroin respectively) while cotton is cellulose — so the shared burnt-hair smell reflects shared protein chemistry, and the wool/silk difference reflects two different proteins

4. A student reasons: 'Wool keeps me warm, so wool must be giving off heat like a small heater.' Trace the flaw in this reasoning chain and pick the corrected version.

- (A) The reasoning is correct; wool fibres undergo a slow chemical reaction that releases warmth
- (B) The flaw is the fibre type; only dark-coloured wool gives off heat by absorbing it from the room
- (C) The flaw is treating insulation as heat production; wool produces no heat — its crimped fibres trap still air, a poor conductor, which slows the escape of the body's own heat
- (D) The flaw is the source of heat; the warmth comes from the dye, not the wool, so undyed wool is not warm

5. Two scarves have equal mass. Scarf 1 is loosely knitted from highly crimped wool; Scarf 2 is the same wool pressed into thin, dense felt. A third scarf, Scarf 3, is smooth silk of the same mass. Ranked WARMEST to COOLEST on a cold day, the order is most likely:

- (A) Scarf 1, then Scarf 2, then Scarf 3 — the lofty crimped wool traps the most still air, compressed felt traps less, and smooth silk traps the least
- (B) Scarf 2, then Scarf 1, then Scarf 3 — denser fabric is always warmer because it blocks more wind

(C) Scarf 3, then Scarf 1, then Scarf 2 — silk is an animal fibre like wool but lighter, so it must be warmest

(D) All three are equally warm, because they have equal mass and warmth depends only on mass

6. A child says, 'If I unravel a wool sweater and a cotton shirt and burn one fibre from each, the safest single test to tell which is which is to see which one shrinks in hot water.' Evaluate this plan.

(A) The shrink test is best, because a single cotton fibre shrinks in hot water faster than wool

(B) Burning is the more reliable single test here: the burnt-hair smell marks wool (protein) and the papery smell marks cotton (cellulose); a 'shrink test' needs the whole garment and warm agitation, not a single fibre, so it is the weaker choice

(C) Neither test works, because wool and cotton behave identically in fire and water

(D) The shrink test is best, because wool dissolves in hot water and disappears, proving it is wool

7. A herder must choose between two highland flocks. Flock A is a fine-wool breed with very few coarse guard hairs; Flock B has long, thick guard hairs and a sparse undercoat. The herder wants maximum YARN SOFTNESS, not maximum total fibre weight. The sounder choice and reason is:

(A) Flock A, because softness comes from a high proportion of fine undercoat with few coarse guard hairs, even if each animal yields less total fibre

(B) Flock B, because more guard hairs mean more wool, and more wool is always softer

(C) Either flock, because softness is decided entirely by scouring and dyeing, not by the breed

(D) Flock B, because long guard hairs can be sorted out and the rest will be softer than Flock A's wool

8. An exam claim reads: 'Sheep in hot lowlands grow extra-thick fleece to shed heat, so India should rear its wool sheep on the coast.' Identify every error in this two-part claim.

(A) Only the first part is wrong; the coast is still the correct rearing region for fine wool

(B) Only the second part is wrong; sheep really do grow thick fleece to shed heat

(C) Both parts are correct, because heat stimulates fleece growth and warm regions give the best wool

(D) Both parts are wrong: thick fleece is grown for warmth in COLD climates (not to shed heat), and India's wool sheep are accordingly reared in cool hill states like Jammu & Kashmir and Himachal Pradesh, not the hot coast

9. Selective breeding has been used for generations to improve wool. A breeder wrongly hopes that, by always breeding the silkiest-fleeced sheep, the flock will eventually 'start producing silk.' The correct judgement is:

- (A) Possible — repeated breeding for silkiness slowly converts the sheep's keratin into silk fibroin
- (B) Possible only if the chosen sheep are also fed mulberry leaves, the silkworm's food
- (C) Impossible, but only because sheep cannot be bred at all for fleece quality
- (D) Impossible by breeding — selective breeding can make a sheep's wool finer, softer and more uniform, but wool is keratin made by the sheep; it can never become silk, which is fibroin spun by a silkworm larva

10. A diagram of a single sheep hair under magnification shows tiny overlapping scales along its surface and a wavy (crimped) shape. Linking these two structural features to two everyday wool facts, the correct pairing is:

- (A) The overlapping scales explain why wool felts and shrinks in hot water with agitation; the crimped wavy shape explains why wool traps air and feels warm and bulky
- (B) The scales explain the warmth and the crimp explains the felting — the two features simply swap roles
- (C) Both features only affect colour, so a smooth straight wool fibre would warm you just as well
- (D) The scales make wool a plant fibre and the crimp makes it an animal fibre

11. The same Pashmina goat breed is reared in two places: high in cold Ladakh and on a warm lowland farm. The Ladakh goats grow a much finer, denser undercoat. The statement that best links climate, animal and fibre warmth is:

- (A) The lowland goats grow the finer undercoat, because heat makes an animal thicken its coat to shed warmth
- (B) Climate cannot affect the undercoat at all; fineness depends only on the dye and scouring used later
- (C) Cold prompts the Ladakh goats to grow a thick, fine undercoat that traps insulating air for the animal, and that same air-trapping fine fibre is exactly what makes Pashmina warm for humans
- (D) The lowland goats begin to grow silk instead of wool, because warmth converts keratin into fibroin

12. A wool mill lists its steps but two are swapped: shearing → SORTING → SCOURING → picking burrs → dyeing → spinning. Identify the swap and why it breaks the process.

- (A) Sorting and scouring are swapped — you must scour (wash out grease and dirt) before sorting, because grades and fine-vs-coarse hairs can only be judged cleanly on washed wool, not on greasy fleece
- (B) Picking burrs and dyeing are swapped — dye must be applied before burrs are removed

(C) Shearing and spinning are swapped — spinning the yarn must come before the sheep is sheared

(D) Nothing is swapped — sorting greasy fleece before washing it is the normal, correct order

13. A factory tries to save time by dyeing the fleece immediately after shearing, before scouring and sorting. Predict the most likely twofold problem.

(A) There is no problem; dyeing first actually cleans the wool and grades it automatically

(B) The grease and dirt block even dye uptake so the colour is patchy, and dyeing before sorting locks coarse and fine hairs into the same dyed batch, so they can no longer be cleanly separated for fine yarn

(C) The only problem is that the dye will smell of burnt hair when the wool is later worn

(D) The wool will turn into silk, because dye applied early changes the fibre's protein

14. In wool processing, two cleaning-related jobs are scouring and picking burrs. A worker insists 'scouring already removes the burrs, so burr-picking is unnecessary.' The accurate correction is:

(A) The worker is right; the warm water of scouring dissolves the burrs completely

(B) Scouring washes out grease, dust and dirt but does not pull out the embedded plant burrs, so a separate burr-picking step is still needed after scouring

(C) The worker is right, because burrs are made of grease and so are washed away with the grease

(D) The worker is wrong, but only because burr-picking must be done before shearing the sheep

15. Order this full wool chain so that each step is only attempted once its prerequisite is met: dyeing, spinning into yarn, shearing, picking burrs, scouring, sorting. The correct sequence is:

(A) shearing → sorting → scouring → dyeing → picking burrs → spinning into yarn

(B) shearing → scouring → sorting → picking burrs → dyeing → spinning into yarn

(C) scouring → shearing → sorting → picking burrs → spinning into yarn → dyeing

(D) shearing → scouring → dyeing → sorting → picking burrs → spinning into yarn

16. A quality auditor finds that the 'sorting' and 'picking burrs' steps were both skipped, but scouring, dyeing and spinning were done. Which TWO defects in the finished yarn are most directly traceable to those skips?

(A) Coarse and fine hairs were spun together (no sorting), and bits of plant burr remain in the yarn (no burr-picking), making the yarn uneven and prickly

(B) The yarn is greasy and dirty, and it cannot take dye, because skipping sorting leaves grease behind

(C) The yarn smells of burnt hair and is flammable, because skipping burr-picking traps spores

(D) The yarn has turned into cotton, because only sorting keeps wool a protein fibre

17. Why must scouring precede BOTH sorting and dyeing, making it a kind of 'gatekeeper' step? Choose the explanation that captures both dependencies.

(A) Scouring adds the colour and the grades at once, so sorting and dyeing are just repeats of scouring

(B) Scouring must come first only for dyeing; sorting can be done equally well on greasy fleece

(C) Scouring must come first only for sorting; dye actually soaks better into greasy, unwashed wool

(D) Scouring removes grease and dirt, and clean wool is required both to judge the fibres correctly during sorting and to absorb dye evenly during dyeing — so both later steps depend on scouring being done first

18. Batch 1 is processed in the correct order (scour, then pick burrs); Batch 2 has burrs picked first on greasy raw fleece, then is scoured. After both are spun into yarn, which batch's yarn is more likely to stay prickly with embedded plant bits, and why?

(A) Batch 1, because scouring before burr-picking glues the burrs permanently into the fibre

(B) Batch 2, because grease and dirt hide the burrs on raw fleece so many are missed, and the later scouring washes out grease but cannot pull out embedded burrs

(C) Neither — the order cannot affect how many burrs remain, since the two steps act independently

(D) Batch 1, because burrs left on during scouring scrub the fibre smooth and remove themselves

19. A sheep yields 4 kg of raw fleece. Scouring removes grease and dirt amounting to 30% of the raw mass. Sorting then discards 10% of the SCOURED wool as too coarse.

Approximately how much sorted fine wool remains?

(A) About 2.5 kg

(B) About 2.8 kg

(C) About 1.6 kg

(D) About 4.0 kg

20. A textbook lists six wool steps and asks which step changes the fibres' APPEARANCE without cleaning them, grading them, or twisting them into yarn. The step is dyeing — but a student picks scouring 'because washing also changes how the wool looks.' The precise correction is:

(A) The student is right, because scouring is the step that adds colour to the wool

(B) Neither dyeing nor scouring fits; sorting is the appearance step because it arranges the fibres

- (C) Both dyeing and scouring fit equally, so the question has two correct answers
- (D) Scouring changes appearance only as a side effect of CLEANING; the question asks for the step whose PURPOSE is appearance without cleaning/grading/spinning, which is dyeing (adding colour)

21. A herder shears Flock X at the onset of winter and identical Flock Y in early summer. Predict which flock suffers and which is unharmed, with the reason.

- (A) Flock X suffers — stripped of its fleece just as the cold arrives, it loses body heat with no insulating coat; Flock Y is unharmed because it does not need the fleece in warm months and regrows it before the next winter
- (B) Flock Y suffers, because summer shearing sunburns the skin and permanently stops wool regrowth
- (C) Neither is affected, because the timing of shearing has no bearing on the sheep's warmth
- (D) Flock X thrives, because removing the fleece in winter makes the sheep grow it back thicker and warmer

22. A student sees that scouring (wool) and reeling (silk) both use HOT water and concludes 'the heat must do the same job in both.' Refute this by naming the different job the heat does in each step.

- (A) The student is right — in both, heat simply sets the dye colour into the fibre
- (B) Heat in both steps only adds the twist that turns loose fibre into finished yarn
- (C) The student is wrong: in scouring the hot water loosens and washes grease and dirt OUT of the raw fleece, whereas in reeling the hot water softens the sericin gum so the single cocoon filament can be unwound — same agent, opposite jobs
- (D) The student is wrong because reeling uses cold water and only scouring uses hot water

23. A care label on a pure-wool blanket reads 'cool wash, do not wring, dry flat.' Explain, in cause-and-effect terms, why these three instructions all guard against the SAME underlying problem.

- (A) All three keep the wool from dissolving, because wool dissolves in hot, moving water
- (B) All three avoid heat and mechanical agitation, which make the scaly wool fibres interlock and felt — cool water limits heat, no wringing limits agitation, and drying flat avoids the stretching and rubbing that also promote felting and distortion
- (C) All three protect the dye, because heat and motion wash the colour out but do not affect the fibre
- (D) All three keep the wool from turning into cotton, which happens whenever wool is heated

24. Two woollen sweaters are washed by mistake: Sweater A in hot water with vigorous scrubbing, Sweater B in hot water but left to soak completely still. Which is more likely to

shrink badly, and why?

- (A) Sweater B, because still hot water penetrates deeper and shrinks wool more than moving water
- (B) Both shrink equally, because heat alone fully determines shrinkage regardless of agitation
- (C) Sweater A, because felting needs BOTH heat and agitation; the scrubbing supplies the agitation that makes the scales interlock, while the still soak of B lacks that agitation
- (D) Neither shrinks, because shrinkage requires cold water, not hot

25. Wool's warmth, its tendency to felt, and its smell when burnt are three different observations. Each points to a property of wool. The correctly matched set is:

- (A) Warmth → protein chemistry; felting → trapped air; burnt-hair smell → overlapping scales
- (B) Warmth → overlapping scales; felting → protein chemistry; burnt-hair smell → trapped air
- (C) All three are caused by the dye, so undyed wool shows none of these properties
- (D) Warmth → crimp traps air; felting → overlapping scales interlock; burnt-hair smell → protein (keratin) chemistry

26. A scientist measures that a wool fabric and a cotton fabric of identical weave and thickness are kept in the same warm room, then both touched. The wool feels warmer to the hand. The best single reason is:

- (A) Wool is at a higher temperature than the cotton even though both sat in the same room
- (B) Wool conducts heat away from the hand more slowly than cotton because its crimped, scaly fibres trap more insulating air, so less body heat is drawn off and it feels warmer
- (C) Wool releases stored heat into the hand because it is an animal fibre
- (D) Cotton feels cooler because it is a synthetic fibre that absorbs heat

27. A child has a wool sweater that has already felted and shrunk after a hot wash. They ask whether soaking it in cold water will 'un-shrink' it back to size. The correct answer is:

- (A) Yes — cold water reverses felting because cold relaxes the scales and lets the fibres slide apart
- (B) Yes — but only if hot water is used again, since heat both shrinks and un-shrinks wool
- (C) No — felting is essentially permanent because the interlocked scales do not unlock by soaking; cold water cannot reverse the matting once it has happened
- (D) No — because the sweater has already turned into cotton and cotton cannot be stretched

28. Why does a thick wool jumper feel warm in cold air but clammy and less effective when it is soaking wet?

- (A) Wet wool is warmer, because the water itself is a better insulator than air
- (B) Water turns the wool protein into cellulose, which cannot trap heat
- (C) Dry, the crimped fibres trap insulating air; when soaked, water fills the air spaces, and water conducts heat far better than air, so the jumper loses its insulating air pockets and feels cold/clammy
- (D) Wet wool feels cold only because the dye washes out and dye is what keeps you warm

29. The silk-moth life cycle is egg → larva → pupa → adult moth. A student is asked to identify both the stage that EATS the most and the single event at which silk is SPUN. The correct pair is:

- (A) The adult moth eats the most, and silk is spun by the moth after it lays its eggs
- (B) The pupa eats the most, and silk is spun at the egg → larva transition as the egg hatches
- (C) The egg eats the most, and silk is spun by the pupa as it becomes a moth
- (D) The larva eats the most (it feeds heavily on mulberry leaves), and silk is spun at the larva → pupa transition when the larva winds its cocoon

30. A documentary narrator says, 'Watch the beautiful moth spin its silk cocoon before it flies away.' A biology teacher objects. The precise correction is:

- (A) The cocoon is spun by the LARVA, not the adult moth; by the time the moth exists, the silk has long been spun and the moth's role is to emerge and lay eggs, not to spin
- (B) The narrator is correct; the adult moth spins the cocoon just before flight
- (C) The cocoon is spun by the pupa as it forms, so neither larva nor moth spins it
- (D) The cocoon is spun by the egg, which is why eggs are wound around the pupa

31. Which single statement about the silk-moth life cycle is FALSE?

- (A) The pupa hatches directly from the egg, and the larva forms later after the cocoon is spun
- (B) The egg hatches into a larva, which is the silkworm that feeds on mulberry leaves
- (C) The larva spins a cocoon around itself as it changes into a pupa
- (D) The adult moth emerges from the cocoon and lays eggs, beginning the cycle again

32. A sericulturist has 1000 cocoons: a few already show exit holes where moths have emerged, while the rest are intact but moths are due within hours. To salvage the most reelable silk overall, the best plan is:

- (A) Wait for every moth to emerge, then reel all 1000 cocoons equally, since emergence does not affect the thread
- (B) At once heat/steam the still-intact cocoons to stop further emergence and keep their filaments unbroken, accepting that the few already-holed cocoons now yield only short broken lengths

(C) Reel the already-holed cocoons first, because an opened cocoon unwinds into one longer thread than an intact one

(D) Place mulberry leaves beside the cocoons so the pupae feed and delay emerging

33. A claim states: 'Feeding silkworms more and fresher mulberry leaves and feeding sheep mulberry leaves will both improve the fibre.' Judge each half of the claim.

(A) Both halves are true, because mulberry leaves improve every animal fibre

(B) Both halves are false, because diet has no effect on any fibre quality

(C) The first half is true (well-fed larvae spin larger, better cocoons with more silk); the second half is false (sheep do not eat mulberry leaves to make wool, and diet cannot turn wool into silk)

(D) The first half is false and the second half is true, because mulberry leaves help sheep but not silkworms

34. At which life-cycle transition does the insect both STOP feeding and BEGIN protecting itself with silk, and what does this imply for silk farmers?

(A) At the larva → pupa transition the fully grown larva stops feeding and spins its silk cocoon for protection, which is exactly when farmers must harvest cocoons to capture the unbroken thread

(B) At the egg → larva transition, so farmers harvest the eggs to get silk

(C) At the pupa → adult-moth transition, so farmers wait for the moth to emerge before harvesting

(D) At the adult-moth → egg transition, so farmers collect silk from the egg-laying moth

35. Raw silk contains fibroin and sericin. A worker accidentally throws away the fibroin and tries to weave with the sericin. Predict the outcome and the correct roles.

(A) Weaving succeeds, because sericin is the strong silk thread and fibroin is the throwaway gum

(B) Weaving succeeds, because both fibroin and sericin are equally good threads

(C) Weaving fails — fibroin is the actual silk filament that becomes cloth, while sericin is only the gummy coating meant to be removed during reeling, so discarding the fibroin discards the usable thread

(D) Weaving fails because both fibroin and sericin are gums and neither can be woven

36. Reeler 1 soaks each cocoon in hot water before unwinding; Reeler 2 skips the soak and pulls the thread off dry. Compare the LENGTH and CONTINUITY of silk each recovers, with the reason.

(A) Reeler 2 recovers a longer thread, because dry silk slides off the cocoon faster than softened silk

(B) Both recover the same single long thread, because hot water is used only to dye silk, not to free it

(C) Reeler 1 recovers one long continuous filament because the hot water softens the sericin gum and releases the thread, while Reeler 2 gets only short broken fragments because the un-softened gum holds the filament fast and snaps it

(D) Neither recovers any thread, because a cocoon is solid gum containing no filament

37. A weaver tries to weave cloth using one single unreeled cocoon filament instead of a reeled multi-filament thread. Predict the result and explain why reeling combines several filaments.

(A) The single fibroin filament is far too fine and weak, so it snaps under the loom's tension and makes no usable cloth; reeling combines several filaments to build a thread strong and thick enough to weave

(B) The cloth weaves perfectly, because one cocoon's filament is already strong enough on its own

(C) It fails because a single filament is far too thick to pass through the loom, so several are combined to thin it

(D) It fails because one cocoon's filament is actually wool and must be blended with silk to weave

38. Reeling and weaving are both silk steps, but only one of them produces fabric. Match each step to what it does:

(A) Reeling interlaces warp and weft threads, while weaving unwinds the cocoon

(B) Reeling unwinds and combines cocoon filaments into thread; weaving then interlaces lengthwise (warp) and crosswise (weft) threads on a loom to make the fabric

(C) Both reeling and weaving unwind cocoons, so neither makes fabric

(D) Reeling boils the cocoon and weaving boils it again, and the heat itself forms the cloth

39. Sericin is removed during reeling rather than left on the finished thread. A worker claims sericin is removed 'because it is poisonous to wear.' Give the accurate reason and refute the claim.

(A) The worker is right; sericin is a toxin that would harm anyone wearing the silk

(B) Sericin is removed because it is the only coloured part and silk must be pure white

(C) Sericin is removed because it is a stiff gum that sticks the filament to itself and to the cocoon, so the thread cannot unwind or weave smoothly while it remains; it is not removed because it is poisonous

(D) Sericin is removed because it makes the filament too thin to weave

40. India produces mulberry silk and also non-mulberry silks. A list reads: tasar, eri, muga, nylon. Which item does NOT belong, and what is the common feature of the rest?

(A) Muga does not belong, because muga is a wool from an Assam goat, not a silk

(B) Eri does not belong, because eri is a synthetic like nylon

(C) Tasar does not belong, because tasar is a grade of sheep's wool

(D) Nylon does not belong — tasar, eri and muga are all natural non-mulberry silks spun by silkworms, whereas nylon is a man-made petroleum-based synthetic

41. Muga silk's natural golden colour is sometimes wrongly explained as coming from boiling the cocoon in golden dye. The accurate statement linking colour, silkworm type and region is:

(A) Muga's golden colour is natural to that particular non-mulberry silkworm and is regionally famous in Assam; boiling softens the gum and does not dye the silk gold

(B) Muga is golden because the cocoon is boiled in a golden dye bath during reeling

(C) All silk is naturally golden and only becomes white after bleaching, so muga is just unbleached silk

(D) Muga is golden because the Assam goat that spins it eats golden mulberry leaves

42. A poster shows the silk journey as: larva feeding on mulberry leaves → reeling → cocoon spinning → weaving. Exactly one step is out of place. Which step, and where does it belong?

(A) Reeling is misplaced — it belongs last, after weaving the cloth

(B) Weaving is misplaced — it belongs first, before the larva feeds

(C) Nothing is misplaced — reeling correctly comes before cocoon spinning

(D) Cocoon spinning is misplaced — you can only reel a cocoon that already exists, so spinning must come before reeling: feeding → cocoon spinning → reeling → weaving

43. A '100% silk' scarf and a '100% cotton' shirt are traced back to their true biological origins. Which single statement is correct for BOTH fibres?

(A) Both began as the fluffy seed hairs from the boll of a plant

(B) Both began as the cocoon filament spun by a silk-moth larva

(C) The silk began as cotton seed hairs and the cotton began as a silk-moth cocoon

(D) The silk began as the continuous fibroin filament a silk-moth larva spun into its cocoon (later reeled), while the cotton began as the seed hairs from a cotton plant's boll — one animal, one plant

44. Four fibres — cotton, wool, silk, jute — are each burnt. Wool and silk smell of burnt hair; cotton and jute smell of burning paper. Using ONLY this burn evidence, the correct grouping is:

(A) Cotton and wool are animal fibres; silk and jute are plant fibres

(B) All four are animal fibres, since the burn test cannot tell plant from animal

(C) Wool and silk are animal (protein) fibres — the burnt-hair smell marks protein — while cotton and jute are plant (cellulose) fibres, marked by the papery smell

(D) Silk and jute are animal fibres; cotton and wool are plant fibres

45. A museum card lists four 'source → fibre → fibre-type' triples. Which single triple is correct on ALL THREE counts?

- (A) Sheep → silk → animal fibre (B) Silk-moth larva → silk → animal fibre
(C) Cotton plant → wool → plant fibre (D) Pashmina goat → silk → animal fibre

46. Tracing a sweater labelled 'pure Pashmina' to its source, then a saree labelled 'pure muga,' the two true origins are:

- (A) Pashmina comes from the fine undercoat of a Himalayan goat (a wool); muga comes from a non-mulberry silkworm in Assam (a silk)
(B) Both come from silkworms; Pashmina is mulberry silk and muga is non-mulberry silk
(C) Both come from goats; muga is a golden goat wool and Pashmina a white goat wool
(D) Pashmina comes from a silkworm cocoon and muga from a Himalayan goat's undercoat

47. A single mulberry cocoon is fully reeled. The length of continuous thread it can yield is best described as:

- (A) only a few centimetres, since each cocoon is small
(B) roughly hundreds to over a thousand metres of one continuous filament
(C) exactly one metre, always the same for every cocoon
(D) shorter than a single human hair, which is why many cocoons are needed

48. Among everyone connected to wool — the shepherd, the sorter, the spinner of finished yarn, and the shopper buying a scoured-and-dyed sweater — who faces the HIGHEST risk of sorter's disease, and why?

- (A) The shopper, because body heat from wearing the sweater activates dormant spores
(B) The sorter, because handling and inhaling dust from raw, unwashed fleece exposes them to anthrax spores that scouring later removes
(C) The spinner, because spinning finished yarn releases the most spores into the air
(D) The shepherd, because drinking the sheep's milk transmits the disease

49. A doctor must name the illness of a raw-fleece wool sorter who has a severe lung infection, and identify its exact causative agent, ruling out look-alike conditions. The precise identification is:

- (A) A fungal lung infection, caught from spores living in the wool dye
(B) Food poisoning, caused by swallowing grease from the raw fleece
(C) A common cold, caused by the cold air of the sorting shed
(D) Sorter's disease — a form of anthrax — caused by inhaled spores of the bacterium *Bacillus anthracis* present in raw fleece

50. The spores that cause sorter's disease are present in raw-fleece dust but gone after scouring. To protect the most workers, a manager must place a single safety 'control point' in the workflow. Where should it go, and why?

- (A) At the finished-sweater packing bench, since the most workers stand there
(B) At the dyeing vats, because dye fumes are what carry the spores

(C) At the raw-fleece sorting stage — disinfect the incoming fleece and give those sorters masks and ventilation, because that is the one stage where spore-laden dust is present and inhaled

(D) At the shop counter, to protect the customers who buy the wool

51. Raw fleece in a sorting shed and a finished sweater in a shop are each tested for anthrax spores. Which carries spores, which does not, and why?

(A) Both carry spores equally, because both are made of the same wool

(B) The sweater carries more, because spores become concentrated as the wool is processed

(C) Neither carries spores, because anthrax can infect only sheep, never wool

(D) The raw fleece carries spores (it is unwashed) but the finished sweater does not, because scouring and full processing wash the spores out before the wool becomes a sweater

52. A safety poster claims 'everyone who handles wool or silk faces an equal anthrax risk.' Correct it by naming the ONE job at genuine risk and saying why the others are not.

(A) The silk reeler is the one at risk, from breathing dye fumes while reeling boiled cocoons

(B) Only the raw-fleece wool sorter is at real risk, because they inhale spores from unwashed fleece; the silk reeler (boiled cocoons), the sweater shopper (scoured goods) and the weaver (processed yarn) all handle spore-free material

(C) The sweater shopper is the one at risk, because body heat from wearing wool activates dormant spores

(D) The claim is right — all four jobs face exactly the same anthrax risk because all touch fibre

53. Spell out the full cause-and-effect chain by which a wool sorter can fall ill, in correct order:

(A) Finished sweater carries spores → the buyer wears it → body heat activates spores → anthrax develops

(B) Raw fleece carries *Bacillus anthracis* spores → sorting stirs up dust → the worker inhales spores → the spores cause anthrax (sorter's disease) in the lungs

(C) Dye contains a virus → fumes are inhaled during dyeing → the virus causes a cold that worsens into anthrax

(D) Scouring water breeds a parasite → the worker drinks it → the parasite causes anthrax in the blood

54. A school sweater is 80% wool / 20% polyester. Mapping each property of the finished sweater to the component that supplies it — warmth, soft feel, added strength, shape-and-wrinkle resistance — the correct mapping is:

(A) Polyester supplies the warmth and the soft feel; the wool merely supplies the colour

(B) Wool supplies the strength and wrinkle-resistance; polyester supplies the warmth

- (C) Polyester supplies all four properties; the wool is only there as cheap filler
- (D) Wool supplies the warmth and soft feel (its crimp traps air), while polyester supplies the added strength, shape-retention and wrinkle-resistance

55. Compared with natural animal fibres like wool and silk, synthetic fibres such as nylon and polyester are best described as:

- (A) man-made from petroleum — durable and wrinkle-resistant, but not biodegradable, and not produced by any plant or animal
- (B) harvested from plants and animals just like wool and silk
- (C) spun by silkworms that feed on petroleum instead of mulberry leaves
- (D) always warmer and softer than wool because they are newer materials

56. An advertiser claims its ordinary mulberry silk is 'cruelty-free — no insect is harmed in normal reeling.' A biologist rejects this. Which correction is accurate?

- (A) The claim is true, because the moth always flies out safely before the cocoon is boiled
- (B) The claim is false, but because the larvae are starved to death, not boiled
- (C) The claim is false, but because the adult moths are killed while laying their eggs
- (D) The claim is false: standard reeling boils the cocoon with the pupa still inside, killing it, precisely so the moth cannot emerge and cut the single continuous thread

57. Pashmina sells for far more per kilogram than ordinary sheep wool. Of the reasons a student lists, the economically sound one is:

- (A) Pashmina costs more because it is dyed gold, like muga silk
- (B) Pashmina costs more because it is a synthetic, and synthetics always cost more than natural fibres
- (C) Each Pashmina goat yields only a little very fine undercoat per year, so the fine fibre is scarce; a scarce and highly desirable fibre commands a high price
- (D) Pashmina and ordinary wool actually cost the same, since all animal fibres are priced equally by weight

58. A manager proposes: 'Rear a fine-wool breed and sort the fleece carefully for softness, and to save time skip scouring so sorters work the greasy raw fleece straight through.' Which goal does this plan secretly fail, and how would you fix it?

- (A) It fails softness, because fine-wool breeds actually grow the coarsest wool
- (B) It fails safety — routing sorters onto unscoured raw fleece leaves them inhaling anthrax spores; fix it by disinfecting the fleece and giving sorters masks and ventilation (clean fibre is softer too)
- (C) It serves both softness and safety perfectly; there is no flaw in the plan
- (D) It fails both goals, because the very act of sorting always spreads spores no matter what

59. A student writes: 'Boiling the cocoon is done only to clean dust off it.' Give the most complete correction, naming all the real purposes.

- (A) The student is fully correct; cleaning the dust is the only reason cocoons are boiled
- (B) Boiling is done to dye the silk a permanent golden colour like muga
- (C) Cocoons are never actually boiled in commercial silk-making
- (D) Boiling chiefly softens the sericin gum so the filament can be reeled, and it kills the pupa so the emerging moth cannot break the long thread; any cleaning is incidental, not the main purpose

60. Two yarns are both sold as 'pure wool.' Yarn 1 weakens when wet, burns papery with grey ash, and never felts. Yarn 2 keeps its strength when wet, chars with a burnt-hair smell, and felts in hot water. Which label is false, and what is each yarn really?

- (A) Yarn 2's label is false — its clues mark it as the cotton one, while Yarn 1 is genuine wool
- (B) Both labels are correct — both sets of clues are normal for wool
- (C) Yarn 1's label is false — every clue (wet-weak, papery burn, grey ash, no felting) marks it as cotton (a plant fibre), while Yarn 2's clues (wet-strong, burnt-hair char, felts) genuinely fit wool, so only Yarn 1 is mislabelled
- (D) Both labels are false — both yarns are really cotton

Solutions — Science (Class 7), Chapter 3: Fibre to Fabric — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	C	21	A	31	A	41	A	51	D
2	A	12	A	22	C	32	B	42	D	52	B
3	D	13	B	23	B	33	C	43	D	53	B
4	C	14	B	24	C	34	A	44	C	54	D
5	A	15	B	25	D	35	C	45	B	55	A
6	B	16	A	26	B	36	C	46	A	56	D
7	A	17	D	27	C	37	A	47	B	57	C
8	D	18	B	28	C	38	B	48	B	58	B
9	D	19	A	29	D	39	C	49	D	59	D
10	A	20	D	30	A	40	D	50	C	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Protein fibres (wool, silk) contain nitrogen and char with a burnt-hair smell; cellulose fibres (cotton, jute) burn with a papery smell and leave soft grey ash; synthetics melt into a bead. P and R both show the burnt-hair signature, so both are animal fibres — exactly two. The idea that grey ash signals protein is wrong: grey papery ash is the cellulose signature, so Q is plant, not animal. Two different threads can certainly both be wool or silk, so 'cannot share the smell' is false. Cotton does NOT smell of burnt hair — it smells like burning paper — so 'both wool and cotton smell of hair' misstates the test that does in fact distinguish them.

2. (A) Silk is a protein fibre: it burns with a burnt-hair smell and is fairly strong even when wet. A papery burn smell and loss of strength when wet are the marks of cotton, a plant cellulose fibre, so the first fabric is mislabelled. The claim that silk weakens and burns papery reverses silk's real behaviour. 'Both are silk because both burn' ignores that the two burn smells are diagnostically different. The second fabric (burnt-hair smell, mats

in hot water) is in fact wool — a protein fibre — not a synthetic; keeping strength when wet is consistent with wool, so calling it synthetic misreads the evidence.

3. (D) The burnt-hair smell is shared because both wool and silk are nitrogen-containing proteins; the difference is that they are DIFFERENT proteins — keratin in wool, fibroin in silk — which is why they feel and behave differently. They are not the identical protein, and dyeing has nothing to do with cotton's chemistry, so the first distractor is wrong twice over. Fibroin is silk, not cotton, and silk is a protein that smells of hair, not the papery cellulose one, so that option scrambles all three assignments. Cotton is cellulose (not protein); its papery smell is the correct, expected result of the test, not an error.

4. (C) Warmth from clothing is insulation, not heat generation. Wool's crimped, scaly fibres trap a layer of still air; air conducts heat poorly, so body heat escapes slowly and you feel warm. Wool itself makes no heat, so the 'small heater' idea and the 'slow chemical reaction' option are both false. Colour/dye does not warm a fibre — undyed creamy fleece insulates just as well — so the dye-based explanations are wrong. The corrected chain is: crimped fibres → trapped still air → slow heat loss → feeling warm.

5. (A) Insulation depends on trapped still air, not on mass. Lofty, crimped wool (Scarf 1) holds the most air; pressing it into felt (Scarf 2) squeezes out much of that air, so it insulates less; smooth silk (Scarf 3) traps the least and is the coolest. Density does not equal warmth — packing the fibres tight reduces the air pockets, so 'denser is always warmer' is wrong. Being an animal fibre does not make silk warm; its smooth fibroin traps little air, so silk is not warmest. Equal mass does not make them equally warm, because structure (air-trapping), not mass, governs insulation.

6. (B) A single fibre cannot meaningfully 'shrink' the way a knitted garment felts; felting needs many scaly wool fibres locking together under heat and agitation, so the shrink idea fails at the single-fibre scale. The burn test on one fibre clearly distinguishes the burnt-hair (protein/wool) smell from the papery (cellulose/cotton) smell, so burning is the better single test. Cotton does not shrink faster than wool — it is wool's scales that cause felting — so that distractor inverts the behaviour. The two fibres do NOT behave identically in fire (different smells), so 'neither works' is false. Wool does not dissolve in hot water; it felts, so the dissolving claim is wrong.

7. (A) Fine, soft yarn comes from breeds rich in fine undercoat hairs and poor in coarse guard hairs (the Merino idea). Flock A fits this directly. More guard hairs do NOT mean softer — they make the fibre coarser — so 'more wool is always softer' confuses quantity with quality. Scouring cleans and dyeing colours; neither changes a coarse hair into a fine one, so softness is not decided by those steps. Sorting can remove some guard hairs from Flock B, but it cannot make Flock B's remaining fibre finer than a breed that is naturally fine; you cannot sort your way past a coarse undercoat, so Flock A still wins.

8. (D) Fleece is insulation against cold, not a heat-shedding device, so a hot lowland is the wrong environment for thick-fleece wool sheep; this makes the first part wrong. Because cold prompts thick fleece, India rears its wool sheep in cool highlands (J&K, Himachal), not the hot coast, so the second part is wrong too — both parts fail. Saying only one part is wrong leaves the other false claim standing. 'Both correct' accepts the inverted biology that heat thickens fleece, which contradicts why a fleece exists.

9. (D) Selective breeding shifts traits within what the animal can already produce — here, finer and more uniform wool — but it cannot make a sheep produce a fibre from a different organism. Wool is keratin from a sheep; silk is fibroin spun by a silkworm larva; no amount of breeding turns one into the other, so the conversion idea is biologically false. Mulberry leaves feed silkworms, not sheep, and diet cannot give a sheep the silk-spinning apparatus, so that option is also wrong. Sheep CAN be bred for fleece quality (that is the whole point of breeds like Merino), so the last option's reason is false even though its 'impossible' verdict is right for the wrong reason.

10. (A) The microscopic scales interlock under heat and agitation, which is exactly why wool felts and shrinks — so scales → felting. The crimp gives the fibre bulk and creates air pockets, so crimp → trapped air → warmth. The swap option assigns each cause to the wrong effect. Neither feature is about colour, and a smooth straight fibre would trap less air and not felt, so the 'colour only' option is wrong on both counts. Scales and crimp are features of an animal protein fibre; they do not make a fibre plant-derived, so the last option is nonsense — wool is animal regardless.

11. (C) The chapter's rule is that cold climates prompt thick fleece. Reared high in cold Ladakh, the Pashmina goats grow a thick, fine undercoat that traps insulating air for the animal's own survival, and that same air-trapping fine fibre is exactly what makes Pashmina warm for people — so the colder rearing yields the finer, warmer wool. The lowland claim inverts the biology (heat does not make an animal thicken its coat 'to shed warmth'). Climate clearly does affect the undercoat, so blaming fineness on dye and scouring (later steps that only clean and colour) is wrong. Warmth never converts a goat's keratin into silk fibroin, so the lowland goats cannot 'grow silk.'

12. (A) Scouring removes grease and dirt; only after washing can you reliably sort the wool by texture, length and colour, so scouring must precede sorting — the listed order swaps them. Dyeing comes AFTER burr-picking (you colour clean, burr-free wool), so claiming dye must precede burr removal reverses the real order. Shearing is the first step and spinning is the last, so swapping them is absurd — you cannot spin yarn before you have the fleece. Sorting greasy, unwashed fleece is exactly the mistake the swap creates, so 'nothing is swapped' is false.

13. (B) Dye soaks evenly only into clean fibre, so dyeing greasy unscoured fleece gives blotchy colour — the first problem. Sorting separates fine from coarse hairs for quality; if you dye before sorting, both hair types end up dyed together in one batch, undermining

the later sorting needed for fine yarn — the second problem. Dyeing does not clean or grade wool (scouring and sorting do that), so the 'no problem' option is false. Dye does not create a burnt-hair smell, and dye cannot transmute keratin into silk fibroin, so the last two options are nonsense.

14. (B) Scouring is a wash that removes greasy and dirty material; burrs are small bits of plant matter caught in the fleece and are picked out by hand in a separate step, so scouring does not eliminate them — the worker is wrong. Warm water does not dissolve woody plant burrs, so the 'dissolves completely' claim is false. Burrs are plant debris, not grease, so they are not washed away with the grease. Burr-picking is done on sheared, scoured fleece, never before shearing (there is no fleece to pick burrs from before shearing), so the last option's reason is wrong even though its 'worker is wrong' verdict is right by accident.

15. (B) You must first remove the fleece (shearing), then wash it (scouring), then grade it (sorting), then remove plant burrs, then add colour (dyeing), and finally twist it into yarn (spinning). The first wrong order sorts before scouring, but you grade clean wool, not greasy fleece. The second wrong order scours before shearing, which is impossible (you cannot wash a fleece still on the sheep) and also dyes after spinning, which misplaces dyeing. The third wrong order dyes before sorting and burr-picking, but colour is added to clean, sorted, burr-free wool, so dyeing belongs near the end.

16. (A) Sorting separates coarse guard hairs from the fine undercoat; skip it and both end up spun together, so the yarn is uneven. Burr-picking removes embedded plant matter; skip it and burrs remain, making the yarn prickly — these are the two direct defects. Grease and poor dye uptake are caused by skipping SCOURING, which was done here, so that distractor blames the wrong step. Burr-picking has nothing to do with smell, flammability or anthrax spores, so that option is irrelevant. No step turns wool into cotton; wool stays keratin regardless, so the last option is impossible.

17. (D) Clean wool is the prerequisite for two separate later jobs: sorting (you can only grade texture, length and colour reliably on washed fibre) and dyeing (dye soaks evenly only into clean fibre). So scouring gates both — that is the full explanation. Scouring neither dyes nor grades wool, so it is not a 'repeat' of those steps. Sorting greasy fleece is unreliable, so the claim that sorting works equally well on greasy wool is false. Dye does NOT soak better into greasy wool — grease repels even dyeing — so the last option inverts the chemistry.

18. (B) Batch 2 keeps more prickly plant bits. Burrs are easiest to see and remove on clean wool; on Batch 2's greasy, dirty raw fleece the grease and dirt hide them, so many are missed during the early burr-picking. The later scouring washes out grease and dirt but does not pull embedded plant burrs free, so those missed burrs stay in and end up spun into the yarn. Batch 1 is the correctly ordered one, so it is not the prickly batch; scouring first does not 'glue' burrs in. The order clearly does matter (clean-then-pick vs pick-then-

clean differ), so 'neither' is false. Burrs are inert plant debris, not self-removing scrubbers, so the last option is wrong.

19. (A) Scouring removes 30% of 4 kg, leaving 70% of 4 = 2.8 kg of scoured wool. Sorting then discards 10% of that 2.8 kg, leaving 90% of 2.8 = 2.52 kg $\approx$ 2.5 kg. The 2.8 kg figure stops after scouring and forgets the sorting loss. The 1.6 kg figure wrongly subtracts both 30% and 10% from the ORIGINAL 4 kg ($4 - 1.2 - 0.4 = 2.4$, and over-subtracting more), mishandling the second percentage. The 4.0 kg figure ignores both losses, treating raw fleece as if no mass were removed.

20. (D) Dyeing's purpose is purely to change appearance (colour); it does not clean, grade or spin — so it is the unique fit. Scouring's PURPOSE is cleaning; any change in look is incidental, so it fails the 'without cleaning' condition. Scouring does not add colour — dyeing does — so the student's reason is also factually wrong. Sorting arranges fibres by quality, not by appearance in the colour sense, so it is not the appearance step. Only dyeing satisfies all the constraints, so there are not two correct answers.

21. (A) Flock X is sheared just as winter begins, so it loses its insulating fleece exactly when it most needs to slow heat loss, and the animals suffer cold stress; Flock Y is sheared in early summer, when it does not need the fleece for warmth, and it regrows a fresh fleece before the next winter — so Y is unharmed. Summer shearing does not 'sunburn the skin and stop regrowth' — sheep regrow fleece year after year — so that option is false. Timing clearly does matter to the sheep's warmth, so 'neither affected' is wrong. Removing the fleece in winter does not make it grow back thicker; it just leaves the animal cold, so the 'thrives' option is the opposite of the truth.

22. (C) The student is wrong: the same agent (hot water) does opposite jobs. In scouring, the hot water loosens and washes grease and dirt OUT of the raw fleece — material is removed. In reeling, the hot water softens the sericin gum that binds the cocoon so the single filament can be unwound and drawn off — a thread is released. So heat cleans in one step and frees a thread in the other. Neither step sets dye colour (dyeing is a separate step), so 'sets the dye' is false. The twist that makes yarn comes from spinning/throwing, not from either of these, so 'only adds the twist' is wrong. Reeling does use hot water (that is its whole point), so 'reeling uses cold water' is false.

23. (B) Felting is caused by heat plus agitation locking the fibre scales together; cool washing, not wringing, and drying flat each remove one source of heat or agitation, so all three target the single felting/shrinking problem. Wool does not DISSOLVE in hot water — it felts — so the 'dissolving' explanation is wrong. The instructions protect the fibre structure, not merely the dye; heat and agitation deform the fibre itself, so the dye-only reason is too narrow and partly false. Heating wool does not convert it into cotton (keratin cannot become cellulose), so the last option is impossible.

24. (C) Wool felting requires heat AND mechanical agitation; the scrubbing of Sweater A provides the agitation that drives the scales to interlock, so A shrinks more. A still soak (B) has heat but little agitation, so it felts less. The claim that still water 'penetrates deeper' to shrink more ignores that agitation, not penetration depth, drives felting. Heat alone is not sufficient — agitation matters — so 'shrink equally' is wrong. Shrinkage is promoted by hot water, not cold, so 'requires cold water' is the opposite of the truth.

25. (D) Each observation maps to a distinct property: warmth comes from crimp trapping air; felting comes from the surface scales interlocking; the burnt-hair smell comes from wool being a nitrogen-rich protein. The first wrong set scrambles these (e.g., it blames warmth on protein chemistry and the burn smell on scales). The second wrong set scrambles them differently. Dye explains none of these — undyed creamy fleece is still warm, still felts, and still smells of burnt hair when burnt — so the dye option is false.

26. (B) Both fabrics reach room temperature, so they start equally warm; the difference is how fast each conducts the hand's heat away. Wool's air-trapping structure conducts slowly, so the hand loses less heat and wool feels warmer. The fabrics are not at different temperatures — they shared the room — so the 'higher temperature' option is false. Wool stores no heat to release; being an animal fibre does not make it a heater. Cotton is a natural plant fibre, not a synthetic, so calling it synthetic and heat-absorbing is doubly wrong.

27. (C) Felting interlocks the fibre scales irreversibly; soaking (hot or cold) does not unlock them, so the shrinkage is essentially permanent — the sweater cannot be returned to size. Cold water does not 'relax' the scales to slide apart; that is the opposite of how felting works, so the first 'yes' is wrong. Heat does not un-shrink wool — it promotes more felting — so the second 'yes' is false. The wool has not turned into cotton; it is still keratin, just matted, so the last option's reason is incorrect even though 'no' is right.

28. (C) Wool insulates by trapping still air; when water replaces that air, and since water conducts heat much better than air, the insulation collapses and the jumper feels clammy and cold. Water is a far better heat conductor than air, so 'water is a better insulator' is exactly backwards. Soaking does not convert keratin into cellulose, so the chemistry-change option is impossible. Dye does not provide warmth, and wetting does not necessarily wash out dye, so the dye explanation is wrong.

29. (D) The larva (silkworm) is the voracious feeding stage, eating mulberry leaves to grow, and it is the larva that spins silk as it transforms into a pupa — so feeding and spinning both centre on the larva. The adult moth does not feed heavily nor spin silk (it lays eggs), so that pair is wrong. The pupa is the resting transformation stage inside the cocoon; it neither eats most nor spins at hatching, so that pair is wrong. The egg does not eat, and the pupa does not spin during emergence (the cocoon is already made), so the last pair is wrong.

30. (A) Silk is spun by the larva (caterpillar) as it prepares to become a pupa; the cocoon already exists before the adult moth forms, so the moth never spins it — the narrator is wrong. The pupa does not spin the cocoon; it is the stage that rests inside the cocoon the LARVA already spun, so 'pupa spins it' misplaces the event. An egg cannot spin silk and is not 'wound around the pupa,' so that option is biologically meaningless. Only the larva-spins-the-cocoon statement is correct.

31. (A) The true order is egg → larva → pupa → adult moth. The false statement reverses larva and pupa, claiming the pupa hatches from the egg and the larva forms later — that is backwards, so it is the correct choice for a 'false' question. It is true that the egg hatches into a larva (silkworm) that eats mulberry leaves, that the larva spins the cocoon as it becomes a pupa, and that the adult moth emerges and lays eggs — so those three are valid and not the answer.

32. (B) A cocoon is one long continuous filament; a moth that chews its way out cuts it into short, less useful pieces. The few already-holed cocoons are therefore lost causes (short fragments only). To salvage the most reelable silk overall, heat/steam the still-intact cocoons at once to stop further emergence and keep their filaments whole — that protects the bulk of the batch. Waiting for every moth to emerge would ruin all the intact ones too, so 'reel all equally' is wrong. An opened cocoon does not unwind into a longer thread — it is already broken — so reeling the holed ones first is mistaken. Pupae inside cocoons do not feed, and adult moths eat no mulberry leaves, so leaving leaves out cannot delay emergence.

33. (C) Mulberry leaves are the silkworm larva's food; abundant fresh leaves let the larva grow and spin a better cocoon, so the first half is true. Sheep do not eat mulberry leaves to improve wool, and no diet converts wool keratin into silk, so the second half is false — making the split judgement correct. 'Both true' wrongly assumes mulberry helps every fibre animal. 'Both false' denies that the silkworm's diet matters, which contradicts well-known sericulture practice. The last option inverts both halves, since mulberry helps silkworms (not sheep).

34. (A) The mature larva stops eating and spins its protective silk cocoon as it becomes a pupa; this is the moment the continuous filament exists intact, so farmers harvest cocoons then. The egg → larva transition is hatching, not silk-spinning, so harvesting eggs yields no silk. Waiting for the moth to emerge breaks the thread (the moth chews out), so harvesting after emergence ruins the silk. The moth → egg phase is reproduction, not silk production, so collecting from the egg-laying moth gives no usable thread.

35. (C) Fibroin is the silk filament that is woven into fabric; sericin is the sticky gum coating that is softened and removed during reeling. Throwing away the fibroin throws away the thread, so weaving with only sericin fails. The claim that sericin is the strong thread and fibroin the throwaway gum reverses their roles. They are not equally good

threads — only fibroin is the woven fibre, so 'both equally good' is false. Fibroin is a fibre, not a gum, so 'both are gums, neither weavable' wrongly classifies fibroin.

36. (C) Reeler 1's hot-water soak softens the sericin gum, releasing the single filament so it unwinds as one long continuous thread. Reeler 2 skips the soak, so the un-softened gum holds the filament fast; pulling it dry snaps it into short broken fragments instead of one strand — so the two reelers recover very different lengths and continuity. Dry silk does not slide off faster (the gum makes it stick), so Reeler 2 does not get a longer thread. Hot water in reeling frees the thread by softening gum, not by dyeing, so 'used only to dye' is wrong. A cocoon is one long fibroin filament coated in sericin gum, so 'no filament, solid gum' denies the very thread silk is made of.

37. (A) A single fibroin filament is extremely fine and weak, so on a loom it snaps under tension and makes no usable cloth — which is exactly why reeling combines several filaments into one thread strong and thick enough to weave. So weaving from one bare filament fails; it is not 'already strong enough,' so that option is wrong. The single filament is too thin, not too thick, so 'combine to thin it' inverts the problem. Reeling joins silk filaments to each other, not silk to wool — there is no wool in reeling — so the last option is wrong.

38. (B) Reeling unwinds the cocoon's filaments and combines them into thread; weaving interlaces warp and weft threads on a loom to make cloth — so reeling makes thread, weaving makes fabric. The first distractor swaps the two definitions. Weaving does not unwind cocoons, so 'both unwind, neither makes fabric' is wrong — weaving clearly makes fabric. Boiling belongs to reeling (to soften gum), not to weaving, and heat does not 'form cloth' by itself, so the last option is nonsense.

39. (C) Sericin is a natural gum; while it coats the filament it binds the thread to itself, so the thread cannot be unwound or woven smoothly — that mechanical reason is why it is softened and removed in reeling. It is not removed for toxicity, so the 'poisonous' claim is false. Sericin is not silk's colour source, so 'only coloured part' is wrong. Removing the gum does not make the filament too thin — the gum is an extra coating, not the strand's body — so the thinness reason is also wrong.

40. (D) Tasar, eri and muga are all natural silks produced by silkworms other than the common mulberry one, so they share a natural-silk origin; nylon is a synthetic made from petroleum, so nylon is the odd one out. Muga is a non-mulberry SILK (famous from Assam), not a goat wool, so 'muga is wool' is wrong. Eri is a natural silk, not a synthetic, so grouping eri with nylon is false. Tasar is a silk, not sheep's wool, so 'tasar is a wool grade' is wrong.

41. (A) Different silkworm types give different natural silks; muga's golden sheen is a natural property of its silkworm and is the pride of Assam. Boiling in reeling softens sericin gum — it does not dye the silk gold — so the dye-bath explanation is false. Most silk

(mulberry) is naturally whitish, not golden, so 'all silk is golden until bleached' is wrong. Muga is spun by a silkworm, not a goat, and goats do not spin silk, so the 'Assam goat eating golden leaves' option is biologically impossible.

42. (D) The poster puts reeling before cocoon spinning, but you can only reel a cocoon that already exists — so cocoon spinning is the misplaced step and must come before reeling, giving feeding → cocoon spinning → reeling → weaving. Reeling is not the last step (weaving is), so 'reeling belongs after weaving' is wrong. Weaving is the final step, not the first, so it does not belong before feeding. And it is not true that 'nothing is misplaced': reeling cannot precede the cocoon it unwinds, so option C is exactly the error.

43. (D) The two fibres have different origins: silk began as the continuous fibroin filament a silk-moth larva spun into its cocoon (later reeled into thread), while cotton began as the fluffy seed hairs from the boll of a cotton plant — one animal, one plant. So the silk did not begin as plant seed hairs and the cotton did not begin as a larva's cocoon, which rules out 'both seed hairs' and 'both cocoons,' as well as the swapped option that gives silk a cotton origin and cotton a cocoon origin. Only the description that keeps each fibre with its true source is correct on BOTH counts.

44. (C) The burn evidence does the classifying: a burnt-hair smell marks a nitrogen-rich protein (animal) fibre, while a burning-paper smell marks a cellulose (plant) fibre. Wool and silk smell of burnt hair, so both are animal (protein) fibres; cotton and jute smell papery, so both are plant (cellulose) fibres — the correct split. Pairing cotton with wool as 'animal' contradicts cotton's papery smell. The burn test absolutely can separate plant from animal (that is its whole point), so 'all four animal' is false. Calling silk and jute 'animal' miscalls jute (papery → plant) and miscalls wool a plant fibre, so it scrambles three of the four.

45. (B) Only 'silk-moth larva → silk → animal fibre' is right on all three counts: the larva is the source, silk is the fibre, and silk is an animal (protein) fibre. 'Sheep → silk' fails the fibre (sheep give wool, not silk). 'Cotton plant → wool → plant fibre' fails the fibre (a cotton plant gives cotton, not wool), even though 'plant fibre' is the right type. 'Pashmina goat → silk' fails the fibre (the goat gives fine wool, not silk), so it too is wrong despite naming an animal source. The triple only holds for the larva/silk/animal line.

46. (A) Pashmina is a fine goat WOOL (animal fibre from a goat's undercoat); muga is a golden SILK spun by a non-mulberry silkworm in Assam — different fibre classes from different animals. They do not both come from silkworms — Pashmina is goat wool — so that option is wrong. They do not both come from goats — muga is silkworm silk — so that option is wrong. The last option swaps the two sources entirely, which is exactly backwards.

47. (B) A cocoon is built from one extremely long filament, often hundreds to over a thousand metres; this enormous continuous length is why reeling recovers one long thread

and why one cocoon is so valuable. 'A few centimetres' wildly underestimates the length. The length is not a fixed one metre — it varies and is far longer. The filament is vastly longer than a human hair, so 'shorter than a hair' is the opposite of the truth; many cocoons are combined for STRENGTH, not because each is tiny.

48. (B) Sorter's disease (anthrax) is contracted by inhaling spores in raw, unwashed fleece dust; the sorter handles exactly that material, so the sorter is at highest risk. Once wool is scoured, the spores are removed, so a shopper's finished sweater carries essentially none, and body heat does not 'activate' spores. Spinning happens after scouring, so the spore-laden dust is already gone — the spinner is low risk. The recognised route is inhalation of raw-fleece spores, not drinking milk, so the shepherd-via-milk option misstates transmission.

49. (D) A raw-fleece sorter who inhales spores develops sorter's disease — a form of anthrax — caused by the bacterium *Bacillus anthracis* whose spores ride on raw, unwashed fleece. The other options misname the agent or route: dye carries no fungal spores, so 'fungal infection from the dye' is wrong; the illness comes from inhaled spores, not swallowed grease, so 'food poisoning' misses the route; and a common cold is a mild virus from person-to-person contact, not from cold shed air. Only the anthrax answer matches both the inhaled-spore route and the bacterial cause.

50. (C) The spores live in raw-fleece dust and are gone after scouring, so the single point where the hazard actually exists is the raw-fleece sorting stage. Placing the control point there — disinfecting incoming fleece and giving those sorters masks and ventilation — guards the one stage where spore-laden dust is present and inhaled, protecting the workers genuinely at risk. The packing bench handles finished, spore-free goods, so siting it there guards no one from anthrax despite the crowd. Dye fumes do not carry the spores, so the dyeing vats are the wrong locus. The shop counter handles processed wool and protects customers who face no real risk, missing the route entirely.

51. (D) The raw fleece is unwashed and so carries the anthrax spores; the finished shop sweater does not, because scouring and full processing wash the spores out before the wool becomes a garment. So the two test very differently — the difference is processing, not chance. They do not carry spores 'equally,' since one is washed and one is not. Spores are removed, not concentrated, by processing, so the sweater does not carry 'more.' And anthrax CAN infect humans (that is the whole occupational hazard), so 'neither, because it infects only sheep' is false. Only the raw-fleece-yes / sweater-no answer fits.

52. (B) The poster's 'equal risk for everyone' is wrong: only the raw-fleece wool sorter is at genuine anthrax risk, because they alone inhale spores from unwashed fleece. The silk reeler handles boiled (spore-free) cocoons, the sweater shopper handles scoured finished goods, and the weaver handles processed yarn — none of them meets live raw-fleece spores. So the reeler-via-dye-fumes line is false (dye fumes carry no anthrax), the shopper-

via-wearing line is false (finished wool is spore-free), and the 'all equal' line is the very over-generalisation being corrected. Only the sorter line holds.

53. (B) The correct chain starts with spores in raw fleece, dust raised during sorting, inhalation, and then anthrax in the lungs. A finished sweater is scoured and spore-free, and body heat does not 'activate' spores, so that chain is false at every link. Dye does not contain an anthrax-causing virus, and a cold does not 'worsen into anthrax,' so that chain is wrong. Anthrax here is bacterial and inhaled, not a water-borne parasite drunk in scouring water, so the last chain misidentifies both the agent and the route.

54. (D) Each component supplies what it is good at: the wool's crimped, scaly fibres trap air, giving warmth and a soft feel, while the tough synthetic polyester adds strength, shape-retention and wrinkle-resistance. So the correct mapping gives warmth and soft feel to wool and durability/shape to polyester. Polyester does not supply the warmth (its smooth fibre traps little air), so the option that hands warmth and feel to polyester is wrong, as is the one that gives polyester the warmth and wool the strength. And the wool is not mere 'filler' nor the polyester an all-rounder — each contributes a distinct property, so 'polyester supplies everything' is false.

55. (A) Nylon and polyester are manufactured chemically from petroleum; they are durable and wrinkle-resistant but do not biodegrade and are not grown by any organism. They are not harvested from plants or animals — that describes natural fibres — so the first distractor is wrong. No silkworm eats petroleum or spins synthetic fibre, so that option is biologically absurd. Synthetics are not 'always warmer and softer' than wool — wool's air-trapping crimp often makes it warmer — so newness does not guarantee superiority.

56. (D) The 'cruelty-free' claim is false: standard reeling boils the cocoon with the pupa still inside, killing it, precisely so the moth cannot emerge and chew through the single continuous thread — the killing is built into ordinary reeling, not avoided. So the moth does not 'fly out safely first,' which kills option A. The stage killed is the pupa by boiling, not larvae by starvation (larvae are well fed to spin good cocoons) and not adult moths during egg-laying — so the two 'false, but for the wrong reason' options misname the stage and method. Only the boiled-pupa account is right.

57. (C) Each Pashmina goat yields only a little very fine undercoat per year, so the fine fibre is genuinely scarce; a fibre that is both scarce and highly desirable commands a high price — that is the sound economic reason it costs far more than ordinary sheep wool. Pashmina is not dyed gold (that confuses it with muga silk), so the 'golden dye' reason is wrong. Pashmina is a natural animal wool, not a synthetic, so 'synthetics always cost more' mis-sources it. And animal fibres are clearly not all priced equally by weight — quality and scarcity drive price — so the 'same cost' option is false.

58. (B) The plan's fine breed plus careful sorting genuinely serve softness, but skipping scouring to route sorters onto greasy raw fleece secretly fails the SAFETY goal: unscoured

raw fleece still carries anthrax spores, and unprotected sorters inhale them. The fix is to scour/disinfect the fleece and give sorters masks and ventilation — and since clean wool is also softer, this helps both goals. It does not fail softness (fine breeds give the finest wool, not the coarsest), so that option is wrong. It is not flawless either, so 'serves both perfectly' is false. And sorting does not 'always spread spores no matter what' — protected sorting of disinfected fleece is safe — so the 'fails both' option overstates the risk.

59. (D) Boiling has two real purposes: softening the sericin gum for reeling and killing the pupa so the continuous thread stays whole; cleaning is incidental, so the student is wrong. Cleaning is not the only reason, so 'fully correct' is false. Boiling does not dye the silk — muga's gold is natural, not from boiling — so the dye claim is wrong. Cocoons ARE boiled in commercial silk production, so 'never boiled' is false.

60. (C) Test the clues against each yarn. Yarn 1 weakens when wet, burns papery with grey ash, and never felts — every one of those is a cotton (plant cellulose) trait, so its 'pure wool' label is false; Yarn 1 is cotton. Yarn 2 keeps its strength when wet, chars with a burnt-hair smell, and felts in hot water — all genuine wool (protein) traits — so Yarn 2's label is honest. Therefore only Yarn 1 is mislabelled. Saying Yarn 2 is the cotton one reverses the evidence; 'both are wool' ignores Yarn 1's papery, grey-ash, non-felting clues; and 'both are cotton' wrongly discards Yarn 2's burnt-hair, felting, wet-strong wool signature.